

Physics:

Q. No.	Subject	Correct Answer
1	Physics	30 N
2	Physics	19.2 N
3	Physics	6×10^4 N
4	Physics	5 m/s^2
5	Physics	10 kg
6	Physics	112.5 N
7	Physics	$m\sqrt{(g^2+a^2)} - ma$
8	Physics	$a \sin\theta$
9	Physics	5 kg each
10	Physics	6:5:3
11	Physics	$g/2$
12	Physics	10 N
13	Physics	50 m/s
14	Physics	Zero
15	Physics	3.2 m/s
16	Physics	3 bullets/sec
17	Physics	I true, II false
18	Physics	492 N
19	Physics	Zero
20	Physics	2 s
21	Physics	9 N
22	Physics	3:1
23	Physics	1 N
24	Physics	$Mg(L-y)/L$

25	Physics	$\tan^{-1}(a/g)$ in direction
26	Physics	$1/\sqrt{m}$
27	Physics	Zero
28	Physics	Acceleration=0
29	Physics	10 m/s
30	Physics	30 N
31	Physics	1
32	Physics	(18, -1.5)
33	Physics	1.2 N
34	Physics	2 m/s^2
35	Physics	Both true, reason correct
36	Physics	$2Mg \sin\theta / 3$
37	Physics	105 N
38	Physics	39 m/s
39	Physics	2.5 m/s^2
40	Physics	$g/\sqrt{(x^2-1)}$
41	Physics	25 m/s^2
42	Physics	7500 N
43	Physics	$t=1 \text{ s}$
44	Physics	500 N
45	Physics	30 N
46	Chemistry	$\text{XeF}_2, \text{XeF}_4, \text{XeF}_6, \text{PCl}_5$
47	Chemistry	$9\sigma, 9\pi$
48	Chemistry	NH_4Cl
49	Chemistry	SF_4
50	Chemistry	N in NO_3^-
51	Chemistry	$\text{ClO}_2^-, \text{SF}_4, \text{XeF}_4$

52	Chemistry	c & d
53	Chemistry	Both correct
54	Chemistry	XeF ₂ linear, XeF ₄ square planar, XeO ₃ pyramidal, XeOF ₄ square pyramidal
55	Chemistry	-1,0,+1
56	Chemistry	H ₂ O
57	Chemistry	SO ₄ ²⁻
58	Chemistry	Diagram 1
59	Chemistry	Point C
60	Chemistry	N ₂ H ₄
61	Chemistry	N ₂
62	Chemistry	i,ii,iv
63	Chemistry	I alone
64	Chemistry	s-p overlap
65	Chemistry	lp-lp > lp-bp > bp-bp
66	Chemistry	Only d
67	Chemistry	Structure 4
68	Chemistry	a,c,d
69	Chemistry	Option 3 incorrect
70	Chemistry	NH ₃ > NH ₄ ⁺ > NH ₂ ⁻
71	Chemistry	sp, sp ² , sp ³
72	Chemistry	SeF ₄ & CH ₄ same shape (wrong)
73	Chemistry	H ₂ O angle larger than CH ₄ (false)
74	Chemistry	d _{z²} & d _{x²-y²}
75	Chemistry	180°
76	Chemistry	sp
77	Chemistry	OF ₂ < Cl ₂ O < H ₂ O < NH ₃

78	Chemistry	b,c,d
79	Chemistry	π bond allows free rotation (false)
80	Chemistry	3,1
81	Chemistry	Statement C incorrect
82	Chemistry	sp^2
83	Chemistry	Linear
84	Chemistry	8
85	Chemistry	3
86	Chemistry	C_2H_6
87	Chemistry	NF_5
88	Chemistry	7
89	Chemistry	sp^2
90	Chemistry	12 lone e^- , 4 bond pairs
91	Botany	Puccinia=wheat rust, Ustilago=smuts, etc.
92	Botany	Mode of locomotion
93	Botany	Nostoc & Anabaena
94	Botany	Morphological characters
95	Botany	Different cell wall structure
96	Botany	Chitin
97	Botany	All
98	Botany	Four
99	Botany	Protista
100	Botany	Slime mould spores dispersed by air
101	Botany	Citrus canker
102	Botany	Gonyaulax
103	Botany	A-3, B-4, C-2, D-1
104	Botany	A-2, B-1, C-4, D-3

107	Botany	One longitudinal, one transverse
108	Botany	Flagellated protozoans
109	Botany	Phycomycetes
110	Botany	Cyanobacteria
111	Zoology	Chrysophytes heterotrophic (wrong)
112	Zoology	Zoospores, aplanospores, conidia
113	Zoology	Gonyaulax
114	Zoology	Albugo–Parasite–Phycomycetes
115	Zoology	A-ii, B-i, C-iv, D-iii
116	Zoology	I true, II false
117	Zoology	Monera
118	Zoology	Pigments identical to green plants
119	Zoology	Both true, reason not explanation
120	Zoology	Protista
121	Zoology	Both false
122	Zoology	Presence of plastids
123	Zoology	Only statement 1 correct
124	Zoology	Ascomycetes & Basidiomycetes
125	Zoology	Cyanobacteria, A= N_2 fixing cell
126	Zoology	Many organisms didn't fit
127	Zoology	A=Cell wall, B=DNA, C=Cell membrane
128	Zoology	Chrysophytes
129	Zoology	i,ii,iii incorrect; iv correct
130	Zoology	ii,iii,v
131	Zoology	Millions of years
132	Zoology	Deuteromycetes
133	Zoology	Protein rich pellicle

134	Zoology	Both asexual & sexual
135	Zoology	Both incorrect
136	Zoology	In vertebrates X→vertebral column, Y embryonic
137	Zoology	X closed, Y closed, Z bony
138	Zoology	Ophiuira & Asterias=Echinodermata; Octopus & Pila=Mollusca
139	Zoology	Both true, R explains A
140	Zoology	Both true, R explains A
141	Zoology	b & c only
142	Zoology	Sexes separate, fertilization internal
143	Zoology	Statement I incorrect, II correct
144	Zoology	A-v, B-iv, C-iii, D-ii, E-i, F-vi
145	Zoology	Catla
146	Zoology	Only A correct
147	Zoology	Endoskeleton consists of feathers... (wrong)
148	Zoology	I incorrect, II correct
149	Zoology	Locomotion
150	Zoology	a,c,d,e
151	Zoology	A-3, B-2

152	Zoology	Hydra (nerve cells present, brain absent)
153	Zoology	Two openings (mouth & anus)
154	Zoology	Aquatic annelids (parapodia)
155	Zoology	a-1, b-3, c-2 (coelom types)
156	Zoology	Ascaris, Ancylostoma, Wuchereria (dioecious)
157	Zoology	Roundworms, organ system, unisexual, internal fertilization, endoparasite
158	Zoology	Two incorrect statements (B and D)

160	Zoology	a & b only (polyps asexual, medusae sexual)
161	Zoology	Cnidoblast – Coelenterata
162	Zoology	Ascidia (notochord only in larval tail)
163	Zoology	Cartilaginous cranium & vertebral column
164	Zoology	Both true, reason correct (Limulus = Arthropoda, living fossil)
165	Zoology	Post-anal tail (distinguishing chordate feature)
166	Zoology	I & II only (dorsal, hollow nerve cord)
167	Zoology	Ectoderm & Endoderm (ctenophores diploblastic)
168	Zoology	Apis (honey), Limulus (living fossil)
169	Zoology	I & III only (jointed appendages, Malpighian tubules)
170	Zoology	Porifera → Cnidaria → Platyhelminthes → Annelida → Arthropoda → Mollusca → Echinodermata → Chordata
171	Zoology	Dorsal to nerve cord, ventral to gut
172	Zoology	Sea-hare (odd member, mollusc not coelenterate)
173	Zoology	Mantle & visceral hump
174	Zoology	Cross, internal fertilization (sponges)
175	Zoology	Pila (open) & Pristis (closed)
176	Zoology	Mesoderm & bilateral symmetry (diploblastic lack)
177	Zoology	B, C, D (moist skin, head+trunk, 3-chambered heart)
178	Zoology	Ostrich – flightless bird
179	Zoology	Chondrichthyes – cartilaginous endoskeleton
180	Zoology	Petromyzon (jawless fish, ammocoetes larva)

Q. No.	Question (from paper)	Options (A–D)	Correct Answer	Reasoning
1	Three blocks connected, pulled with $T_3=60\text{ N}$. Find T_1 .	A. 10 N B. 20 N C. 30 N D. 40 N	C (30 N)	Distribution of force across masses.
2	Two blocks on inclined planes, tension T_1 .	Options given	B (19.2 N)	Net force balance on inclines.
3	30 wagons pulled by engine $6\times 10^5\text{ N}$. Coupling force?	A. 2×10^4 B. 4×10^4	C ($6\times 10^4\text{ N}$)	Force per wagon coupling.

		C. 6×10^4 D. 8×10^4		
4	Pulley with $m_1=10$ kg, $m_2=6$ kg. Acceleration?	A. 10 B. 5 C. 2.5 D. 1.5	B (5 m/s²)	Net force = $(m_1-m_2)g / (m_1+m_2)$.
5	Force $(6i+8j)$ N gives acceleration $(2i+3j)$. Mass?	A. 10 B. 2 C. 70 D. 4	A (10 kg)	$F = ma$ vector relation.
6	Two blocks 4 kg & 8 kg, upward force 200 N. Tension at P?	Options given	C (112.5 N)	Net force distribution.
7	Body hanging in car, tension in string.	Options given	B ($m\sqrt{(g^2+a^2)} - ma$)	Vector addition of accelerations.
8	Wedge in elevator, block acceleration wrt wedge.	Options given	B ($a \sin\theta$)	Relative acceleration in non-inertial frame.
9	10 kg block suspended by two spring balances.	Options given	B (5 kg each)	Load equally shared.
10	Lift moving downward, tensions T_1, T_2, T_3 .	Options given	C (6:5:3)	Variation with acceleration phases.
11	Find acceleration of mass m (smooth surface, force $2mg$).	Options given	A ($g/2$)	Net force = $2mg$, acceleration = $g/2$.
12	From graph find magnitude of negative force.	Options given	B (10 N)	Area under curve = impulse.
13	Net force 10 kg-wt on 10 kg body, velocity after 5 s.	Options given	D (50 m/s)	Acceleration = 10, $v=at=50$.
14	Force-time curve, momentum acquired 0–8 s.	Options given	A (Zero)	Positive & negative impulses cancel.
15	$F=2t+6$, velocity at $t=2$ s, $m=5$ kg.	Options given	C (3.2 m/s)	Integrate acceleration.
16	Machine gun fires 40 g bullet at 1200 m/s, max force 144 N. Bullets/sec?	Options given	D (3)	Force = rate $\times$ momentum.
17	Elevator statements.	Options given	B (I true, II false)	Uniform speed balanced; downward acceleration reduces floor force.
18	Person 60 kg in descending lift $a=1.8$ m/s ² . Apparent weight?	Options given	A (492 N)	Apparent weight = $m(g-a)$.
19	Three blocks $m, 2m, 3m$ pulled upward at constant speed. Net force on $2m$?	Options given	B (Zero)	Constant speed $\rightarrow$ net force zero.
20	Retarding force 50 N on 10 kg body, $v=10$ m/s. Time to stop?	Options given	A (2 s)	$a=F/m=5$, $t=v/a=2$.
21	Three blocks A,B,C (1,8,27 kg), $T_3=36$ N. Find T_2 .	Options given	C (9 N)	Ratio of tensions.

22	1 kg mass lifted with $a=4.9$, lowered with $a=4.9$. Ratio of tensions?	Options given	D (3:1)	Tension differs with direction.
23	Gravels dropped on conveyor belt 0.5 kg/s, $v=2$ m/s. Extra force?	Options given	A (1 N)	$F=\dot{m}v=0.5\times 2=1$.
24	Chain of mass M , length L , tension at distance y .	Options given	B ($Mg(L-y)/L$)	Linear distribution of weight.
25	Sphere suspended in accelerating car. Inclination of string?	Options given	A ($\tan^{-1}(a/g)$ in direction of motion)	Balance of forces.
26	Velocity acquired by mass m over distance d under constant force.	Options given	C ($1/\sqrt{m}$)	$v \propto 1/\sqrt{m}$.
27	Position-time graph of 2 kg body, impulse at $t=2$ s.	Options given	D (Zero)	No change in momentum.
28	Essential characteristic of equilibrium.	Options given	B (Acceleration = 0)	Equilibrium means no net acceleration.
29	Force vs time graph, velocity after 12 s ($m=18$ kg).	Options given	B (10 m/s)	Impulse/mass = velocity.
30	Cricketer catches ball 150 g, $v=20$ m/s, $t=0.1$ s. Force?	Options given	C (30 N)	$F=\Delta p/\Delta t=3/0.1=30$.
31	Mechanical advantage of single fixed pulley.	Options given	A (1)	Fixed pulley only changes direction.
32	Body 1 kg, $u=(3i+4j)$, $F=(2i-3j)$, find coordinates after 3 s.	Options given	A (18, -1.5)	Use equations of motion.
33	Bullet 0.04 kg, $v=90$ m/s, stopped in 3 s. Resistive force?	Options given	B (1.2 N)	$F=\Delta p/\Delta t=3.6/3=1.2$.
34	Three masses 1,6,3 kg connected, acceleration?	Options given	B (2 m/s^2)	Net force/mass.
35	Assertion: Cement floor injury > sand. Reason: Impulse greater.	Options given	A (Both true, reason correct)	Cement stops faster $\rightarrow$ greater force.
36	System smooth, tension in string.	Options given	B ($2Mg \sin\theta / 3$)	Derived from force balance.
37	Force exerted by 5 kg block in lift $a=5 \text{ m/s}^2$.	Options given	C (105 N)	Apparent weight = $m(g+a)$.
38	Force on 16 kg body, $F=75$ N for 5 s, then 0. Velocity at 10 s?	Options given	D (39 m/s)	$v=at=75/16\times 5\approx 23.4$, but circled ~ 39 .
39	2 kg block struck by water jet $\dot{m}=1$ kg/s, $v=5$ m/s. Acceleration?	Options given	D (2.5 m/s^2)	$F=\dot{m}v=5$ N, $a=F/m=2.5$.
40	Body on smooth incline, horizontal acceleration.	Options given	D ($g/\sqrt{(x^2-1)}$)	Derived from pseudo force balance.

41	Rocket 200 kg, $\dot{m}=50$ kg/s, $v=100$ m/s. Initial acceleration?	Options given	C (25 m/s²)	Thrust= $\dot{m}v=5000$ N, $a=F/m=25$.
42	Helicopter 1000 kg, $a=15$, crew+passenger=300 kg. Force on floor?	Options given	C (7500 N)	Apparent weight = $m(g+a)$.
43	Impulse $I = 20t^2 - 40t$. At what time is change in momentum minimum?	A. $t=2$ s B. $t=1$ s C. $t=\frac{1}{2}$ s D. $t=\frac{3}{2}$ s	B (t=1 s)	Differentiate I wrt t $\rightarrow$ $dI/dt=40t-40$. Minimum at $t=1$ s.
44	Cricket ball 250 g, $v=10$ m/s, returns with same speed in 0.01 s. Force on bat?	A. 25 N B. 50 N C. 250 N D. 500 N	D (500 N)	$\Delta p=0.25 \times 20=5$ Ns, $F=\Delta p/\Delta t=5/0.01=500$ N.
45	Force 12 N gives acceleration 4 m/s ² . Force for 10 m/s ² ?	A. 15 N B. 20 N C. 25 N D. 30 N	D (30 N)	Force 12 N gives acceleration 4 m/s ² . Force for 10 m/s ² ?

Chemistry:

Q. No.	Question (from paper)	Options (A–D)	Correct Answer	Reasoning
46	Hypervalent species (XeF_2 , XeF_4 , XeF_6 , PCl_5).	A. a,b,c,d B. b,c,d C. a,b,c D. c,d,e	A (a,b,c,d)	All listed are hypervalent (expanded octet).
47	Number of σ and π bonds in $\text{C}_2(\text{CN})_4$.	A. 4,4 B. 7,7 C. 9,9 D. 6,6	C (9,9)	Each CN contributes triple bonds; total count = 9σ , 9π .
48	Contains coordinate + covalent bond.	A. BaCl_2 B. NH_4Cl C. HCl D. NaCl	B (NH_4Cl)	NH_4^+ formed by coordinate bond, Cl^- ionic.
49	Central atom not sp^3 hybridized.	A. SF_4 B. BF_4^- C. NH_4^+ D. CH_4	A (SF_4)	SF_4 has sp^3d hybridization.
50	Hybridization of P in PO_4^{3-} same as:	A. I in ICl_4^- B. S in SO_3 C. N in NO_3^- D. S in SO_2	C (N in NO_3^-)	Both sp^2 hybridized.

51	Same hybridization, different shapes.	A. NO_2^- , NO_2 , NO_2^+ B. ClO_2^- , SF_4 , XeF_4 C. NH_3 , H_2O , OF_2 D. SO_4^{2-} , PO_4^{3-} , ClO_4^-	B (ClO_2^-, SF_4, XeF_4)	All sp^3d but different geometries.
52	Correct statements about PCl_5 .	Options given	C (c & d)	Axial bonds longer; solid PCl_5 exists as $[\text{PCl}_4]^+[\text{PCl}_6]^-$.
53	sp hybridization facts.	A. Both correct B. I correct, II incorrect C. Both incorrect D. I incorrect, II correct	A (Both correct)	sp = diagonal hybridization; orbitals 180° .
54	Match Xe compounds with structures.	Options given	2 (a-ii, b-iv, c-iii, d-i)	XeF_2 linear, XeF_4 square planar, XeO_3 pyramidal, XeOF_4 square pyramidal.
55	Formal charges in ozone (O_3).	A. $-1, +1, 0$ B. $0, 0, -1$ C. $-1, 0, +1$ D. $0, +1, -1$	C ($-1, 0, +1$)	Resonance structure distribution.
56	Smallest bond angle.	A. H_2O B. CO_2 C. NH_3 D. BeCl_2	A (H_2O)	Lone pairs reduce angle to $\sim 104.5^\circ$.
57	Ratio of π bonds 1:2.	A. CO_3^{2-} B. SO_4^{2-} C. SO_3 D. ClO_3^-	B (SO_4^{2-})	π bonds distributed 1:2.
58	No bond overlapping diagram.	Options given	1	Opposite phase s orbitals cancel.
59	Stable state of H_2 molecule curve.	A. A B. B C. C D. D	C	Minimum energy at equilibrium bond length.
60	Longest N–N bond.	A. N_2H_4 B. N_2 C. N_2F_2 D. All equal	A (N_2H_4)	Single bond longest.
61	Maximum bond enthalpy.	A. N_2 B. O_2 C. F_2 D. HF	A (N_2)	Triple bond highest enthalpy.
62	Octet rule inadequate for:	A. (i),(ii) B. (i),(ii),(iv) C. (i),(ii),(iii),(iv) D. (ii),(iii),(iv)	B (i,ii,iv)	Expanded octet, geometry, odd electron molecules.

63	Bond overlap statements.	A. I & II B. II & III C. I alone D. II alone	C (I alone)	σ bond from s-s overlap; π bond from lateral overlap, not axial.
64	Overlap in HCl.	A. s-s B. p-p C. s-d D. s-p	D (s-p)	H 1s overlaps Cl 3p.
65	Correct repulsion order.	A. lp-lp > lp-bp > bp-bp B. lp-lp > bp-bp > lp-bp C. bp-bp > lp-bp > lp-lp D. lp-bp > bp-bp > lp-lp	A	Lone pair repulsion strongest.
66	Incorrect overlaps.	A. a&b B. b&d C. Only d D. Only a	C (Only d)	s+py cannot form π bond.
67	Stable structure of XeO_2F_2 (VSEPR).	Options given	4	Bent geometry with two O double bonds, two F single bonds.
68	Change in hybridization.	A. a,b B. a,c C. a,c,d D. Only a	C (a,c,d)	$\text{BF}_3 \rightarrow \text{NH}_3$, $\text{SbF}_5 \rightarrow [\text{SbF}_6]^-$, $\text{SiF}_4 \rightarrow [\text{SiF}_6]^{2-}$ involve hybridization change.
69	Incorrect bond strength order.	Options given	3	$2s-2p > 2p-2p > 3p-3p > 3s-3p$ is wrong.
70	Bond angle order NH_3 , NH_2^- , NH_4^+ .	Options given	3 ($\text{NH}_3 > \text{NH}_4^+ > \text{NH}_2^-$)	Lone pairs reduce angle in NH_2^- .
71	Hybridization of N in NO_2^+ , NO_3^- , NH_4^+ .	Options given	3 (sp, sp^2, sp^3)	NO_2^+ linear sp, NO_3^- trigonal sp^2 , NH_4^+ tetrahedral sp^3 .
72	Wrong statement.	Options given	3 (SeF_4 and CH_4 same shape)	SeF_4 seesaw, CH_4 tetrahedral.

73	False statement among CH ₄ , NH ₃ , H ₂ O bond angles.	Options given	3 (H₂O bond angle larger than CH₄)	Actually H ₂ O smaller (104.5° vs 109.5°).
74	d orbitals in sp ³ d ² hybridization.	Options given	2 (d_z² & d_x²-y²)	Octahedral hybridization uses these.
75	Angle between s–p overlap.	Options given	3 (180°)	Linear overlap.
76	Hybridization of C in C ₃ O ₂ .	Options given	2 (sp)	Carbon in cumulene structure is sp.
77	Correct bond angle order.	Options given	3 (OF₂ < Cl₂O < H₂O < NH₃)	Lone pairs reduce angles accordingly.
78	Species not having identical geometry (I ₃ [–] , CO ₂ , NO ₃ [–] , SO ₂ , XeF ₄ , SF ₆).	A. (b)&(c) only B. (b),(c)&(d) C. (a),(c)&(d) D. (a) only	B (b,c,d)	NO ₃ [–] trigonal planar vs SO ₂ bent; XeF ₄ square planar vs SF ₆ octahedral; NO ₂ bent vs NO ₃ [–] trigonal planar.
79	Incorrect statement about π bonds.	A. π bond doesn't affect shape B. π bonds present with σ C. π bond allows free rotation D. π bond less stable than σ	C (π bond allows free rotation)	π bonds restrict rotation due to electron cloud overlap.
80	Max number of 180° angles in sp ³ d ² vs sp ³ d hybridisation.	A. 3,3 B. 3,1 C. 1,3 D. 3,0	B (3,1)	Octahedral (sp ³ d ²) has 3 linear pairs; trigonal bipyramidal (sp ³ d) has 1.
81	Incorrect statement about hybridisation.	A. Not possible for isolated atoms B. Number of hybrid orbitals = orbitals combined C. Only	C	Empty orbitals can also hybridize; statement is wrong.

		half/full orbitals participate D. Not a real physical process		
82	Hybrid state of N atom (context: NO_2^- etc.).	A. sp^2 B. sp^3 C. sp^3d^2 D. sp	A (sp^2)	Nitrogen in NO_2^- is sp^2 hybridized.
83	Shape of I_3^- .	A. Linear B. Bent C. Pyramidal D. See-saw	A (Linear)	I_3^- has 3 atoms in line, with lone pairs on central I.
84	Number of $109^\circ 28'$ angles in CCl_4 .	A. 2 B. 4 C. 6 D. 8	D (8)	Tetrahedral geometry has 8 bond angles of 109.5° .
85	Number of C atoms in sp^2 hybrid state in $\text{CH}\equiv\text{CH}-\text{CH}_3$.	A. 9 B. 8 C. 3 D. 2	C (3)	Two carbons in triple bond are sp, one in CH_3 is sp^3 , middle carbon is sp^2 .
86	Molecule with maximum C–C bond length.	A. C_2H_2 B. C_2H_4 C. C_2H_6 D. $\text{C}_2\text{H}_2\text{Br}_2$	C (C_2H_6)	Single bond longest; double/triple shorter.
87	Species that does not exist.	A. BF_3 B. NF_3 C. PF_5 D. NF_5	D (NF_5)	Nitrogen cannot expand octet to form NF_5 .
88	Covalency of Cl in second excitation.	A. 2 B. 5 C. 3 D. 7	D (7)	In excited state, Cl can expand octet to covalency 7.
89	Hybridisation of Cl in ClO_3^- .	A. sp B. sp^2 C. sp^3 D. None	B (sp^2)	ClO_3^- has trigonal pyramidal geometry, sp^2 hybridization.
90	Lone pairs and bond pairs in XeF_4 .	A. 2,2 B. 14,4 C. 12,4 D. 12,2	C (12,4)	Xe has 8 valence e^- + 4 bonds = 12 lone electrons (6 pairs) + 4 bond pairs.

Biology:

Biology – Botany (Q91–110, Q141–150, Q151–160) → Plant physiology, ecology, genetics.

Q. No.	Question (from paper)	Options (A–D)	Correct Answer	Reasoning
91	Match Puccinia, Ustilago, Saprophytes, Albugo with traits.	A. i ii iii iv B. ii iii iv i C. iii iv i ii D. iii ii iv i	D (iii, ii, iv, i)	Puccinia = wheat rust, Ustilago = smuts, Saprophytes = dead organic matter, Albugo = parasitic on mustard.
92	Which is not Whittaker's criterion?	A. Mode of nutrition B. Reproduction C. Phylogenetic relationship D. Mode of locomotion	D (Mode of locomotion)	Whittaker used nutrition, reproduction, phylogeny, not locomotion.
93	Nitrogen-fixing cyanobacteria are:	A. Rhizobium & Nostoc B. Nostoc & Anabaena C. Anabaena & Rhizobium D. Methanogens & Anabaena	B (Nostoc & Anabaena)	Both are cyanobacteria with heterocysts for nitrogen fixation.
94	Aristotle classified plants into herbs, shrubs, trees based on:	A. Anatomical B. Physiological C. Morphological D. Reproductive	C (Morphological)	Classification was based on external morphology.

95	Archaeobacteria survive harsh habitats because of:	A. Cell wall B. Oxygen requirement C. Different cell wall structure D. All	C (Different cell wall structure)	Unique cell wall composition allows survival in extremes.
96	Major component of fungal cell wall:	A. Chitin B. Peptidoglycan C. Cellulose D. Hemicellulose	A (Chitin)	Fungal cell walls are chitinous.
97	Heterotrophic bacteria:	A. Some pathogenic B. Abundant in nature C. Nitrogen fixation, blooms D. All	D (All)	They include pathogens, decomposers, nitrogen fixers.
98	How many have aseptate mycelium (list given)?	A. Four B. Three C. Two D. Five	A (Four)	Rhizopus, Mucor, Albugo, etc. have aseptate mycelium.
99	Boundaries not well defined in which kingdom?	A. Fungi B. Protista C. Plantae D. Animalia	B (Protista)	Protista overlaps with plants, animals, fungi.
100	Identify correct statement.	A. Slime mould spores dispersed by air B. Diatoms are prokaryotes C. Fungi are autotrophs D. All	A (Spores dispersed by air)	Diatoms are eukaryotes, fungi heterotrophic.
101	Plant disease caused by bacteria?	A. Cholera B. Typhoid C. Citrus canker D. Smut	C (Citrus canker)	Bacterial disease in plants.
102	Which is a dinoflagellate?	A. Desmid B. Trypanosoma C. Gonyaulax D. Euglena	C (Gonyaulax)	Dinoflagellate causing red tides.

103	Match groups (Dinoflagellates, Diatoms, Slime molds, Mycoplasma).	Options given	3 (A-3, B-4, C-2, D-1)	Dinoflagellates = red tides, Diatoms = siliceous shells, Slime molds = plasmodium stage, Mycoplasma = no cell wall.
104	Match fungi groups.	Options given	2 (A-2, B-1, C-4, D-3)	Phycomycetes = bread mould, Ascomycetes = yeast, Basidiomycetes = puffballs, Deuteromycetes = Alternaria.
107	Dinoflagellates flagella arrangement.	A. Both longitudinal B. One longitudinal, one transverse C. Both transverse D. Both (1)&(3)	B (One longitudinal, one transverse)	Characteristic flagellar arrangement.
108	Sleeping sickness protozoan group.	A. Ciliated B. Flagellated C. Sporozoans D. Amoeboid	B (Flagellated)	Caused by Trypanosoma, a flagellated protozoan.
109	Coenocytic mycelium characteristic of:	A. Phycomycetes B. Ascomycetes C. Basidiomycetes D. Deuteromycetes	A (Phycomycetes)	Rhizopus, Mucor have coenocytic mycelium.
110	Specialized heterocysts present in:	A. Archaeobacteria B. Rhizobium C. Dinoflagellates D. Cyanobacteria	D (Cyanobacteria)	Heterocysts fix nitrogen in cyanobacteria.

Q. No.	Question (from paper)	Options (A–D)	Correct Answer	Reasoning
111	Select the wrong statement. (1) Chrysophytes are heterotrophic. (2) Diatomaceous earth is formed by the cell walls of diatoms. (3) Desmids are known as golden algae. (4) Diatoms are microscopic and float passively in water.	A. Chrysophytes are heterotrophic B. Diatomaceous earth from diatoms C. Desmids = golden algae D. Diatoms float passively	A (Chrysophytes are heterotrophic)	Chrysophytes (like diatoms, desmids) are primarily autotrophic, not heterotrophic. The other statements are correct.
112	The asexual reproduction of fungi may occur by means of – (1) Zoospores, zygospore and ascospores (2) Aplanospores, ascospores and zygospores (3) Zoospores, aplanospores and conidia (4) Aplanospores, zygospores and basidiospores	A. Zoospores, zygospore, ascospores B. Aplanospores, ascospores, zygospores C. Zoospores, aplanospores, conidia D. Aplanospores, zygospores, basidiospores	C (Zoospores, aplanospores and conidia)	Fungi reproduce asexually by motile spores (zoospores), non-motile spores (aplanospores), and conidia. Zygospores, ascospores, basidiospores are sexual spores.

113	<i>Vegetative reproduction by fragmentation is common in</i>	A. Agaricus B. Yeast C. Euglena D. Gonyaulax	D (Gonyaulax)	Fragmentation is typical in algae/protists like Gonyaulax, not fungi like Agaricus or Yeast.
114	<i>Which one of the following matches is correct?</i>	A. Albugo — Parasite — Phycomycetes B. Alternaria — Sexual reproduction absent — Basidiomycetes C. Agaricus — Parasite fungus — Ascomycetes D. Neurospora — Used in genetic work — Basidiomycetes	A (Albugo — Parasite — Phycomycetes)	Albugo is a parasitic fungus under Phycomycetes.
115	<i>Match the following (Symbionts, Penicillium, Morels, Mucor)</i>	A. Symbionts → ii. Mycorrhiza B. Penicillium → i. Conidia C. Morels → iv. Edible fungi D. Mucor → iii. Bread mould	Option 1 (A-ii, B-i, C-iv, D-iii)	Correct matching of fungal examples with features.
116	<i>Statement I: Marine amoeboid protozoans have silica shells. Statement II: Gullet is present in Trypanosoma.</i>	A. I false, II true B. I true, II false C. Both correct D. Both false	B (I true, II false)	Marine amoeboids (radiolarians) have silica shells; Trypanosoma lacks gullet.

117	Which kingdom includes producers, consumers, decomposers (Whittaker)?	A. Plantae B. Fungi C. Protista D. Monera	D (Monera)	Monera includes autotrophs, heterotrophs, and decomposers.
118	Which statement is true?	A. Euglenoids bear two flagella B. Euglena dies in darkness C. Pigments of Euglena identical to green plants D. Thermoacidophiles in marshy areas	C (Pigments identical to green plants)	Euglena has chlorophyll a & b like green plants.
119	Assertion: Majority of Euglenoids are freshwater. Reason: All unicellular eukaryotes under Protista.	A. Both true, reason correct explanation B. Both true, reason not explanation C. Assertion true, reason false D. Both false	B (Both true, reason not explanation)	Euglenoids are freshwater; Protista includes unicellular eukaryotes but not the explanation for habitat.
120	Kingdom forming link with plants, animals, fungi	A. Monera B. Protista C. Fungi D. Plantae	B (Protista)	Protista bridges unicellular forms with plants, animals, fungi.
121	Statement I: Mycoplasma is archaebacteria. Statement II: Golden algae are dinoflagellates.	A. Both true B. I true, II false C. I false, II true D. Both false	D (Both false)	Mycoplasma is not archaebacteria (it's Mollicutes), golden algae are Chrysophytes not dinoflagellates.
122	Which is not a characteristic of fungi?	A. Cell wall of chitin B. Heterotrophic nutrition C.	C (Presence of plastids)	Fungi lack plastids; they are heterotrophic with

		Presence of plastids D. Reproduction by spores		chitin walls and spore reproduction.
123	<i>Statement–1: In slime moulds during favourable conditions, the plasmodium differentiates and fruiting bodies bearing spores at their tips. Statement–2: Ascospores are endogenous and basidiospores are exogenous asexual spores.</i>	A. Both correct B. Both incorrect C. Only statement 1 correct D. Only statement 2 correct	C (Only statement 1 correct)	Slime moulds form fruiting bodies with spores. Ascospores and basidiospores are sexual spores, not asexual.
124	<i>Dikaryophase is a specific property of:</i>	A. Basidiomycetes & Deuteromycetes B. Ascomycetes & Basidiomycetes C. Ascomycetes & Phycomycetes D. Almost all fungi	B (Ascomycetes & Basidiomycetes)	Dikaryotic stage (n+n) is characteristic of Ascomycetes and Basidiomycetes.
125	<i>Identify the figure and correct info: (chain-like structure, cell 'A')</i>	A. Protistan algae, A = reproductive structure B. Eubacteria, A = photosynthetic structure C. Cyanobacteria, A = N ₂ fixing cell D. Archaeobacteria, A = heterocyst	C (Cyanobacteria, A = N₂ fixing cell)	The figure shows cyanobacteria (e.g., Nostoc/Anabaena) with heterocyst for nitrogen fixation.

126	<i>Two-kingdom system was inadequate because:</i>	A. Plants & animals classified easily B. Easy to understand C. Both (1) & (2) D. Many organisms did not fit plants/animals	D (Many organisms did not fit)	Protists, fungi, bacteria didn't fit neatly into plant/animal categories.
127	<i>Identify A, B, C in bacterial cell figure</i>	A. Cell wall, B. DNA, C. Cell membrane B. DNA, C. Cell wall, etc.	A (A–Cell wall, B–DNA, C–Cell membrane)	Correct identification of bacterial structures.
128	<i>Group with cell walls forming two thin overlapping shells</i>	A. Slime moulds B. Chrysophytes C. Euglenoids D. Dinoflagellates	B (Chrysophytes)	Diatoms (chrysophytes) have silica cell walls forming two overlapping shells (frustules).
129	<i>(i) Basidiomycetes sex organs present (ii) Some deuteromycetes decomposers, many parasites (iii) Basidiospores endogenously formed (iv) In phycomycetes zygospore formed by gamete fusion</i>	A. (i),(ii) correct; (iii),(iv) incorrect B. (i),(ii),(iii) correct; (iv) incorrect C. (i),(ii),(iii) incorrect; (iv) correct D. (i),(ii) incorrect; (iii),(iv) correct	C (i, ii, iii incorrect; iv correct)	Basidiomycetes lack sex organs, basidiospores are exogenous, deuteromycetes mostly parasites. Zygospore in phycomycetes forms by gamete fusion.
130	<i>Recognise the figures (A = Euglena, B = Agaricus) and</i>	(1) (i),(ii),(iv)&(v) (2) (i),(ii)&(v) (3) (ii),(iii)&(v) (4) (ii),(iii),(iv)&(v)	3 (ii, iii & v)	A is Euglena, B is Agaricus. Both are eukaryotes. B is heterotrophic, A is mixotrophic

	<i>find suitable options.</i>			(autotrophic + heterotrophic).
131	<i>Identify the time period taken by diatomaceous earth for its formation.</i>	A. Millions of years B. Billions of years C. Hundreds of years D. Thousands of years	A (Millions of years)	Diatomaceous earth forms over geological timescales from accumulated diatom shells.
132	<i>Points: (A) Body septate & branched, (B) Decomposers of litter, (C) Help in mineral cycling, (D) Only asexual & vegetative phases known.</i>	A. Phycomycetes B. Euglenoids C. Deuteromycetes D. Basidiomycetes	C (Deuteromycetes)	Deuteromycetes (Fungi Imperfecti) are decomposers, only asexual/vegetative phases known.
133	<i>Euglenoids have flexible body because of:</i>	A. Cellulose rich pellicle B. Protein rich pellicle C. Lipid rich pellicle D. Pectin rich pellicle	B (Protein rich pellicle)	Euglenoids lack a cell wall; pellicle made of protein strips gives flexibility.
134	<i>Which type of reproduction is exhibited by Protista?</i>	A. Asexual reproduction B. Sexual reproduction C. Both asexual & sexual reproduction D. A sort of sexual reproduction	C (Both asexual and sexual)	Protists reproduce by binary fission (asexual) and also sexual processes.
135	<i>Assertion: Bacteria mainly reproduce by sexual reproduction via primitive DNA</i>	A. Assertion correct, Reason incorrect B. Both incorrect C. Both correct, Reason explains Assertion	B (Both incorrect)	Bacteria reproduce mainly by binary fission (asexual). DNA transfer (conjugation, transformation,

	<i>transfer. Reason: Under unfavourable conditions bacteria reproduce by fission.</i>	D. Both correct, Reason not explanation		transduction) is not true sexual reproduction.
136	<i>Identify the correct statement w.r.t. structures marked X and Y (Chordate diagram).</i>	A. Both X & Y present in Urochordates B. Both X & Y circulate water for respiration C. In vertebrates X → vertebral column, Y only embryonic D. X may be present in few non-chordates but Y absent	C	Notochord (X) replaced by vertebral column in adults; pharyngeal gill slits (Y) only embryonic in higher chordates.
137	<i>Identify X, Y, Z in table (Cyclostomata, Chondrichthyes, Osteichthyes).</i>	A. X closed, Y closed, Z bony B. X open, Y closed, Z bony C. X open, Y closed, Z cartilaginous D. X closed, Y open, Z cartilaginous	A	Cyclostomata have closed circulation, Chondrichthyes closed, Osteichthyes bony endoskeleton.
138	<i>Identify A, B, C, D with phyla (Ophiuira, Octopus, Pila, Asterias).</i>	A. Ophiuira Mollusca, Octopus Mollusca, etc. B. Ophiuira Echinodermata, Octopus Mollusca, Pila Mollusca, Asterias Echinodermata C. Ophiuira Echinodermata,	B	Ophiuira & Asterias are echinoderms; Octopus & Pila are molluscs.

		Octopus Echinodermata, etc. D. Ophiuira Mollusca, Octopus Echinodermata, etc.		
139	<i>Assertion:</i> <i>Platyhelminthes</i> <i>dorsoventrally</i> <i>flattened.</i> <i>Reason: Mostly</i> <i>endoparasites</i> <i>in animals.</i>	A. Both true, R explains A B. Both true, R not explanation C. A true, R false D. A false, R true	A	Flattened body is characteristic; parasitic nature explains adaptation.
140	<i>Assertion:</i> <i>Arthropods have</i> <i>open</i> <i>circulation.</i> <i>Reason: Large</i> <i>sinuses, lack</i> <i>capillaries.</i>	A. Both true, R explains A B. Both true, R not explanation C. A true, R false D. A false, R true	A	Arthropods have open system due to hemocoel and sinuses.
141	<i>Identify correct</i> <i>set of</i> <i>statements</i> <i>(ctenophores,</i> <i>roundworms,</i> <i>parasites).</i>	A. a & d only B. b, c & e only C. a, b & d only D. b & c only	D (b & c only)	Ctenophores are exclusively marine, reproduce sexually; flame cells are in Platyhelminthes not roundworms.
142	<i>What is true for</i> <i>Reptilia?</i>	A. External ear opening B. Sexes separate, fertilization internal C. Oviparous with external fertilization D. Presence of mammary glands	B	Reptiles have separate sexes, internal fertilization.
143	<i>Statement I: Air</i> <i>bladder in</i>	A. Both incorrect B. I correct, II	C	Pristis (sawfish) lacks air bladder;

	<i>Pristis regulates buoyancy. Statement II: Echinoderms fertilization external, indirect development.</i>	incorrect C. I incorrect, II correct D. Both correct		echinoderms have external fertilization with larval stages.
144	<i>Match the columns (Physalia, Adamsia, Pennatula, Gorgonia, Meandrina, Aurelia).</i>	Options given	2 (A-v, B-iv, C-iii, D-ii, E-i, F-vi)	Physalia = Portuguese man of war, Adamsia = sea anemone, Pennatula = sea pen, Gorgonia = sea fan, Meandrina = brain coral, Aurelia = jellyfish.
145	<i>Terminal mouth is present in:</i>	A. Catla B. Electric ray C. Shark D. Sting ray	A (Catla)	Catla (Indian carp) has terminal mouth; rays/sharks have ventral mouths.
146	<i>Statement A: All female mammals have milk glands. Statement B: All amphibians have scales.</i>	A. Both correct B. Both incorrect C. Only A correct D. Only B correct	C (Only A correct)	Mammals have mammary glands; amphibians lack scales.
147	<i>Which is incorrect about birds?</i>	A. Skin dry, oil gland at tail B. Hind limb scales, modified for locomotion C. Separate sexes, internal fertilization, oviparous, direct development D. Endoskeleton consists of	D	Feathers, beak, claws are integumentary structures, not endoskeleton.

		feathers, scales, beak, claws		
148	<i>Statement I: Annelids, arthropods, molluscs segmented coelomates. Statement II: Coelenterates, ctenophores, echinoderms radial symmetry.</i>	A. Both incorrect B. I correct, II incorrect C. I incorrect, II correct D. Both correct	C	Statement I wrong (molluscs not segmented); Statement II correct (radial symmetry).
149	<i>Comb plates in sea walnut help in:</i>	A. Respiration B. Locomotion	B (Locomotion)	Comb plates (ctenophores) aid in locomotion, not respiration.
150	<i>Select the group of phyla that includes marine species only.</i>	A. (a,b,c,d) B. (a,c,d) C. (a,c) D. (a,c,d,e)	D (a,c,d,e)	Ctenophora, Echinodermata, Cephalochordata, Hemichordata are exclusively marine.
151	<i>Match Scoliodon, Pristis, Catla, Petromyzon.</i>	Options given	1 (A-3, B-2, C-4, D-1)	Correct classification of fish types.
152	<i>In which animal nerve cell present but brain absent?</i>	A. Sponge B. Catla C. Cockroach D. Hydra	D (Hydra)	Hydra has nerve net but no brain.
153	<i>A complete digestive system has:</i>	A. Single opening both mouth/anus B. Two openings mouth & anus C. Single opening mouth only D. Two	B (Two openings, mouth & anus)	Complete digestive tract has separate mouth and anus.

		openings both mouth/anus		
154	<i>Parapodia are locomotory structures found in:</i>	A. Terrestrial annelids B. Aquatic annelids C. Parasitic annelids D. Aquatic arthropods	B (Aquatic annelids)	Parapodia are lateral appendages in polychaetes.
155	<i>Match body cavity types.</i>	Options given	1 (a-1, b-3, c-2)	Coelom lined by mesoderm, pseudocoelom, acoelom.
156	<i>In which option all animals are dioecious?</i>	A. Fasciola, Ascaris, Taenia B. Ascaris, Ancylostoma, Wuchereria C. Ancylostoma, Fasciola, Ascaris D. Taenia, Ascaris, Wuchereria	B (Ascaris, Ancylostoma, Wuchereria)	All are separate sexes.
157	<i>Diagram description (roundworm).</i>	Options given	4 (Roundworms, organ system, unisexual, internal fertilization, endoparasite)	Matches nematode features.
158	<i>Porifera statements (A–D).</i>	Options given	3 (Two incorrect)	Statements B and D are wrong; porifera not all marine/symmetrical, fertilization internal.
160	<i>Coelentrates statements.</i>	Options given	4 (a & b only)	Polyps produce medusae asexually; medusae produce polyps sexually.

161	<i>Identify figure/phylum.</i>	A. Cnidoblast– Coelenterata B. Flame cell– Platyhelminthes C. Lasso cell– Ctenophora D. Tube feet– Echinodermata	A (Cnidoblast– Coelenterata)	Specialized stinging cells in coelenterates.
162	<i>Notochord present only in larval tail.</i>	A. Ascidia B. Branchiostoma C. Lancelet D. All	A (Ascidia)	Urochordates have notochord only in larval tail.
163	<i>Cranium & vertebral column of cyclostomes.</i>	Options given	Cartilaginous	Cyclostomes have cartilaginous skeleton.
164	<i>Assertion: Limulus is Arthropoda. Reason: Living fossil.</i>	Options given	2 (Both true, reason correct explanation)	Limulus is arthropod and called living fossil.
165	<i>Distinguishing feature of chordates.</i>	A. Segmented body B. Post-anal tail C. Paired appendages D. Ventral solid nerve cord	B (Post-anal tail)	Unique chordate feature.
167	<i>Germinal layers in ctenophores.</i>	A. Ectoderm & Mesoderm B. Mesoderm & Endoderm C. Mesoderm only D. Ectoderm & Endoderm	D (Ectoderm & Endoderm)	Diploblastic animals.
168	<i>Correct organism–role match.</i>	Options given	4 (A & D only)	Apis → honey, Limulus → living fossil.

169	<i>Arthropoda statements.</i>	Options given	1 (I & III only)	Jointed appendages, Malpighian tubules.
170	<i>Correct trend of body organisation.</i>	Options given	1 (Porifera → Cnidaria → Platyhelminthes → Annelida → Arthropoda → Mollusca → Echinodermata → Chordata)	Evolutionary complexity order.
171	<i>Notochord location in chordates.</i>	Options given	1 (Dorsal to nerve cord, ventral to gut)	Correct anatomical position.
172	<i>Odd member in Coelenterata.</i>	A. Sea-pen B. Sea-anemone C. Sea-hare D. Sea-fan	C (Sea-hare)	Sea-hare is mollusc, not coelenterate.
173	<i>Mantle cavity is space between.</i>	Options given	2 (Mantle & visceral hump)	Correct anatomical description.
174	<i>Fertilization in sponges.</i>	Options given	2 (Cross, internal)	Sponges have internal cross fertilization.
175	<i>Pair with open & closed circulation.</i>	Options given	1 (Pila, Pristis)	Molluscs open, fishes closed.
176	<i>Diploblastic animals lack.</i>	Options given	3 (Mesoderm & bilateral symmetry)	Diploblastic have ecto & endoderm only, radial symmetry.
177	<i>Amphibians statements.</i>	Options given	3 (B, C, D)	Amphibians moist skin, body head+trunk, 3-chambered heart. Fertilization external,

				development indirect.
178	<i>Aves correct match.</i>	Options given	3 (Ostrich – flightless bird)	Ostrich is flightless.
179	<i>Group correctly matched.</i>	Options given	3 (Chondrichthyes – cartilaginous endoskeleton)	Sharks, rays have cartilaginous skeleton.
180	<i>Jawless fish with ammocoetes larva.</i>	A. Petromyzon B. Pristis C. Pterophyllum D. Both (1)&(2)	A (Petromyzon)	Lamprey lays eggs in freshwater, larva returns to ocean.

? Q43 → B

? Q44 → D

? Q45 → D

? Q46 → A

? 78 → B

? 79 → C

? 80 → B

? 81 → C

? 82 → A

? 83 → A

? 84 → D

? 85 → C

? 86 → C

? 87 → D

? 88 → D

? 89 → B

? 90 → C

? 91 → D

? 92 → D

? 93 → B

? 94 → C

? 95 → C

? 96 → A

? 97 → D

? 98 → A

? 99 → B

? 100 → A

? 101 → C

? 102 → C

? 103 → 3

? 104 → 2

? 107 → B

? 108 → B

? 109 → A

? 110 → D

? Q111 → Option A

? Q112 → Option C

? Q113 → D

? Q114 → A

? Q115 → Option 1 (A-ii, B-i, C-iv, D-iii)

? Q116 → B

? Q117 → D

? Q118 → C

? Q119 → B

☐ Q120 → B

☐ Q121 → D

☐ Q122 → C

☐ Q123 → C

☐ Q124 → B

☐ Q125 → C

☐ Q126 → D

☐ Q127 → A

☐ Q128 → B

☐ Q129 → C

☐ Q130 → Option 3

☐ Q131 → Option A

☐ Q132 → Option C

☐ Q133 → Option B

☐ Q134 → Option C

☐ Q135 → Option B

☐ Q136 → C

☐ Q137 → A

☐ Q138 → B

☐ Q139 → A

☐ Q140 → A

☐ Q141 → D

☐ Q142 → B

☐ Q143 → C

☐ Q144 → 2

☐ Q145 → A

☐ Q146 → C

☐ Q147 → D

? Q148 → C

? Q149 → B

? 150 → D

? 151 → 1

? 152 → D

? 153 → B

? 154 → B

? 155 → 1

? 156 → B

? 157 → 4

? 158 → 3

? 160 → 4

? 161 → A

? 162 → A

? 163 → Cartilaginous

? 164 → 2

? 165 → B

? 166 → 1

? 167 → D

? 168 → 4

? 169 → 1

? 170 → 1

? 171 → 1

? 172 → C

? 173 → 2

? 174 → 2

? 175 → 1

? 176 → 3

☐ 177 → 3

☐ 178 → 3

☐ 179 → 3

☐ 180 → A